```
> restart: with(LinearAlgebra): with(Groebner):

  m := 13;

  twist := - m + 4;

  p := 5;

  dR := 3*(m-0) + twist:
  dS := 3*(m-2) + twist:
  dU := 3*(m-4) + twist:
  dV := 3*(m-6) + twist:
  dW := 3*(m-8) + twist:

  ##

  AA := 2*T^2 + 1*X^2 + 1*Y^2;
  BB := 1*T^2 + 2*X^2 + 1*Y^2;
  CC := 1*T^2 + 1*X^2 + 2*Y^2;

  #

  AT := diff(AA,T):  BT := diff(BB,T):  CT := diff(CC,T):
  AX := diff(AA,X):  BX := diff(BB,X):  CX := diff(CC,X):
  AY := diff(AA,Y):  BY := diff(BB,Y):  CY := diff(CC,Y):

  #

  ATT := diff(AA,T,T):  ATX := diff(AA,T,X):  ATY := diff(AA,T,Y):
  AXX := diff(AA,X,X):  AXY := diff(AA,X,Y):  AYY := diff(AA,Y,Y):
  BTT := diff(BB,T,T):  BTX := diff(BB,T,X):  BTY := diff(BB,T,Y):
  BXX := diff(BB,X,X):  BXY := diff(BB,X,Y):  BYY := diff(BB,Y,Y):
  CTT := diff(CC,T,T):  CTX := diff(CC,T,X):  CTY := diff(CC,T,Y):
  CXX := diff(CC,X,X):  CXY := diff(CC,X,Y):  CYY := diff(CC,Y,Y):

  #

  MatDD := Matrix([[AT,BT,CT],
                   [AX,BX,CX],
                   [AY,BY,CY]]):

  DD := factor(Determinant(MatDD));

  ##
```

```
Ap  := AT*T1 + AX*X1 + AY*Y1:
Bp  := BT*T1 + BX*X1 + BY*Y1:
Cp  := CT*T1 + CX*X1 + CY*Y1:

App  := + AT*T2 + AX*X2 + AY*Y2
     + ATT*T1*T1 + 2*ATX*T1*X1 + 2*ATY*T1*Y1 + AXX*X1*X1 + 2*
AXY*X1*Y1 + AYY*Y1*Y1:

Bpp  := + BT*T2 + BX*X2 + BY*Y2
     + BTT*T1*T1 + 2*BTX*T1*X1 + 2*BTY*T1*Y1 + BXX*X1*X1 + 2*
BXY*X1*Y1 + BYY*Y1*Y1:

Cpp  := + CT*T2 + CX*X2 + CY*Y2
     + CTT*T1*T1 + 2*CTX*T1*X1 + 2*CTY*T1*Y1 + CXX*X1*X1 + 2*
CXY*X1*Y1 + CYY*Y1*Y1:

##

logACp  := Ap/AA - Cp/CC:
logBCp  := Bp/BB - Cp/CC:

logACpp := App/AA - Ap^2/AA^2 - (Cpp/CC - Cp^2/CC^2):
logBCpp := Bpp/BB - Bp^2/BB^2 - (Cpp/CC - Cp^2/CC^2):

##

Carte_xy := {T =1,  X =x, Y =y,
          T1=0, X1=x1, Y1=y1,
          T2=0, X2=x2, Y2=y2}:

xyAA := subs(Carte_xy, AA):
xyBB := subs(Carte_xy, BB):
xyCC := subs(Carte_xy, CC):
xyDD := subs(Carte_xy, DD):

xylogACp := subs(Carte_xy, logACp):
xylogBCp := subs(Carte_xy, logBCp):

xylogACpp := subs(Carte_xy, logACpp):
xylogBCpp := subs(Carte_xy, logBCpp):

##

Carte_tx := {T =t,  X =x,  Y =1,
          T1=t1, X1=x1, Y1=0,
          T2=t2, X2=x2, Y2=0}:

txAA := subs(Carte_tx, AA):
```

```
txBB := subs(Carte_tx, BB):
txCC := subs(Carte_tx, CC):
txDD := subs(Carte_tx, DD):

txlogACp := subs(Carte_tx, logACp):
txlogBCp := subs(Carte_tx, logBCp):

txlogACpp := subs(Carte_tx, logACpp):
txlogBCpp := subs(Carte_tx, logBCpp):

##

Jet_ABC := +  (1/(T*X*Y)^(m-0))
              *
            add(R[m-0-j,j] * logACp^(m-0-j) * logBCp^j *
(logACp*logBCpp-logACpp*logBCp)^0, j=0..m-0)

          +   (1/(T*X*Y)^(m-2))
              *
            add(S[m-3-j,j] * logACp^(m-3-j) * logBCp^j *
(logACp*logBCpp-logACpp*logBCp)^1, j=0..m-3)

          +   (1/(T*X*Y)^(m-4))
              *
            add(U[m-6-j,j] * logACp^(m-6-j) * logBCp^j *
(logACp*logBCpp-logACpp*logBCp)^2, j=0..m-6)

          +   (1/(T*X*Y)^(m-6))
              *
            add(V[m-9-j,j] * logACp^(m-9-j) * logBCp^j *
(logACp*logBCpp-logACpp*logBCp)^3, j=0..m-9)

          +   (1/(T*X*Y)^(m-8))
              *
            add(W[m-12-j,j] * logACp^(m-12-j) * logBCp^j *
(logACp*logBCpp-logACpp*logBCp)^4, j=0..m-12):

##

xyJet_ABC := factor(+  (1/(x*y)^(m-0))
                    *
              add(Rxy[m-0-j,j] * xylogACp^(m-0-j) * xylogBCp^j
* (xylogACp*xylogBCpp-xylogACpp*xylogBCp)^0, j=0..m-0)

            +   (1/(x*y)^(m-2))
                *
              add(Sxy[m-3-j,j] * xylogACp^(m-3-j) * xylogBCp^j
* (xylogACp*xylogBCpp-xylogACpp*xylogBCp)^1, j=0..m-3)
```

```
            +   (1/(x*y)^(m-4))
                *
              add(Uxy[m-6-j,j] * xylogACp^(m-6-j) * xylogBCp^j
* (xylogACp*xylogBCpp-xylogACpp*xylogBCp)^2, j=0..m-6)

            +   (1/(x*y)^(m-6))
                *
              add(Vxy[m-9-j,j] * xylogACp^(m-9-j) * xylogBCp^j
* (xylogACp*xylogBCpp-xylogACpp*xylogBCp)^3, j=0..m-9)

            +   (1/(x*y)^(m-8))
                *
              add(Wxy[m-12-j,j] * xylogACp^(m-12-j) *
xylogBCp^j * (xylogACp*xylogBCpp-xylogACpp*xylogBCp)^4, j=0..
m-12)):

#

txJet_ABC := +  (1/(t*x)^(m-0))
                *
              add(Rtx[m-0-j,j] * txlogACp^(m-0-j) * txlogBCp^j
* (txlogACp*txlogBCpp-txlogACpp*txlogBCp)^0, j=0..m-0)

            +   (1/(t*x)^(m-2))
                *
              add(Stx[m-3-j,j] * txlogACp^(m-3-j) * txlogBCp^j
* (txlogACp*txlogBCpp-txlogACpp*txlogBCp)^1, j=0..m-3)

            +   (1/(t*x)^(m-4))
                *
              add(Utx[m-6-j,j] * txlogACp^(m-6-j) * txlogBCp^j
* (txlogACp*txlogBCpp-txlogACpp*txlogBCp)^2, j=0..m-6)

            +   (1/(t*x)^(m-6))
                *
              add(Vtx[m-9-j,j] * txlogACp^(m-9-j) * txlogBCp^j
* (txlogACp*txlogBCpp-txlogACpp*txlogBCp)^3, j=0..m-9)

            +   (1/(t*x)^(m-8))
                *
              add(Wtx[m-12-j,j] * txlogACp^(m-12-j) *
txlogBCp^j * (txlogACp*txlogBCpp-txlogACpp*txlogBCp)^4, j=0..
m-12):

##

NFxy := proc(P,j,i)
```

```
expand(mtaylor(P,x,j) + mtaylor(P,y,i) - mtaylor(mtaylor(P,y,i),
x,j)):

end proc:

#

NFtx := proc(P,j,i)

expand(mtaylor(P,t,j) + mtaylor(P,x,i) - mtaylor(mtaylor(P,x,i),
t,j)):

end proc:
```

$$m := 13$$
$$twist := -9$$
$$p := 5$$
$$AA := 2\,T^2 + X^2 + Y^2$$
$$BB := T^2 + 2\,X^2 + Y^2$$
$$CC := T^2 + X^2 + 2\,Y^2$$
$$DD := 32\,T\,X\,Y \tag{1}$$

```
> xyJet := mtaylor(

      factor((x*y*xyAA*xyBB*xyCC)^m * (subs(y2=(xyW+x2*y1)/x1,
xyJet_ABC)))

        , [x1,y1,xyW], 100):

#

txJet := mtaylor(

      factor((t*x*txAA*txBB*txCC)^m * (subs(x2=(txW+t2*x1)/t1,
txJet_ABC)))

        , [t1,x1,txW], 100):
```

```
> printlevel := 2:
  for k from 0 to m/3  do
  for j from 0 to m-3*k do

  xyEEEEq[m-j-3*k,j,k] := factor(coeftayl(xyJet, [x1,y1,xyW]=[0,0,
  0], [m-j-3*k,j,k])):

  od: od:

  #

  printlevel := 2:
  for k from 0 to m/3  do
  for j from 0 to m-3*k do

  txEEEEq[m-j-3*k,j,k] := factor(coeftayl(txJet, [t1,x1,txW]=[0,0,
  0], [m-j-3*k,j,k])):

  od: od:
```

```
> printlevel := 2:
  for k from 0 to m/3  do
  for j from 0 to m-3*k do

  xyEEEq[m-j-3*k,j,k] := mtaylor(

                      factor(
                           (1/(x^(m-j-3*k)*y^j*(x*y)^(3*k))) *
  (1/((xyAA*xyBB*xyCC)^(2*k)))
                              *
                           xyEEEEq[m-j-3*k,j,k])

  , [x,y], 1000):

  od: od:

  #

  printlevel := 2:
  for k from 0 to m/3  do
  for j from 0 to m-3*k do

  txEEEq[m-j-3*k,j,k] := mtaylor(
```

```
                    factor(
                          (1/(t^(m-j-3*k)*x^j*(t*x)^(3*k))) *
(1/((txAA*txBB*txCC)^(2*k)))
                           *
                          txEEEEq[m-j-3*k,j,k])

, [t,x], 1000):

od: od:
```

```
> printlevel := 2:
  for k from 0 to m/3  do
  for j from 0 to m-3*k do

  xyEEq[m-j-3*k,j,k] := NFxy(xyEEEq[m-j-3*k,j,k], j,m-j-3*k) mod p:

  od: od:

  #

  printlevel := 2:
  for k from 0 to m/3  do
  for j from 0 to m-3*k do

  txEEq[m-j-3*k,j,k] := NFtx(txEEEq[m-j-3*k,j,k], j,m-j-3*k) mod p:

  od: od:
```

```
> for j from 0 to m-0 do
  Rxy[m-0-j,j] := add(add(cR[m-0-j,j,dR-q-r,q,r]*1^(dR-q-r)*x^q*
  y^r, q=0..dR-r), r=0..dR):
  od:

  for j from 0 to m-3 do
  Sxy[m-3-j,j] := add(add(cS[m-3-j,j,dS-q-r,q,r]*1^(dS-q-r)*x^q*
  y^r, q=0..dS-r), r=0..dS):
  od:

  for j from 0 to m-6 do
  Uxy[m-6-j,j] := add(add(cU[m-6-j,j,dU-q-r,q,r]*1^(dU-q-r)*x^q*
```

```
y^r, q=0..dU-r), r=0..dU):
od:

for j from 0 to m-9 do
Vxy[m-9-j,j] := add(add(cV[m-9-j,j,dV-q-r,q,r]*1^(dV-q-r)*x^q*
y^r, q=0..dV-r), r=0..dV):
od:

for j from 0 to m-12 do
Wxy[m-12-j,j] := add(add(cW[m-12-j,j,dW-q-r,q,r]*1^(dW-q-r)*x^q*
y^r, q=0..dW-r), r=0..dW):
od:

#

for j from 0 to m-0 do
Rtx[m-0-j,j] := add(add(cR[m-0-j,j,dR-q-r,q,r]*t^(dR-q-r)*x^q*
1^r, q=0..dR-r), r=0..dR):
od:

for j from 0 to m-3 do
Stx[m-3-j,j] := add(add(cS[m-3-j,j,dS-q-r,q,r]*t^(dS-q-r)*x^q*
1^r, q=0..dS-r), r=0..dS):
od:

for j from 0 to m-6 do
Utx[m-6-j,j] := add(add(cU[m-6-j,j,dU-q-r,q,r]*t^(dU-q-r)*x^q*
1^r, q=0..dU-r), r=0..dU):
od:

for j from 0 to m-9 do
Vtx[m-9-j,j] := add(add(cV[m-9-j,j,dV-q-r,q,r]*t^(dV-q-r)*x^q*
1^r, q=0..dV-r), r=0..dV):
od:

for j from 0 to m-12 do
Wtx[m-12-j,j] := add(add(cW[m-12-j,j,dW-q-r,q,r]*t^(dW-q-r)*x^q*
1^r, q=0..dW-r), r=0..dW):
od:
```

```
> printlevel := 2:
  for k from 0 to m/3  do
  for j from 0 to m-3*k do
```

```
xyEq[m-j-3*k,j,k] := NFxy(xyEEq[m-j-3*k,j,k], j,m-j-3*k):

od: od:

#

printlevel := 2:
for k from 0 to m/3  do
for j from 0 to m-3*k do

txEq[m-j-3*k,j,k] := NFtx(txEEq[m-j-3*k,j,k], j,m-j-3*k):

od: od:
```

```
> printlevel := 2:
  for k from 0 to m/3  do
  for j from 0 to m-3*k do

  xyEquations[m-j-3*k,j,k] := {coeffs(xyEq[m-j-3*k,j,k], [x,y])}:

  od: od:

  #

  printlevel := 2:
  for k from 0 to m/3  do
  for j from 0 to m-3*k do

  txEquations[m-j-3*k,j,k] := {coeffs(txEq[m-j-3*k,j,k], [t,x])}:

  od: od:
```

```
> xyEquations_Add := {}:

  for k from 0 to m/3  do
  for j from 0 to m-3*k do
```

```
xyEquations_Add := xyEquations_Add union xyEquations[m-j-3*k,j,k]
:

od: od:

Equations_xy := xyEquations_Add:

#

txEquations_Add := {}:

for k from 0 to m/3   do
for j from 0 to m-3*k do

txEquations_Add := txEquations_Add union txEquations[m-j-3*k,j,k]
:

od: od:

Equations_tx := txEquations_Add:
```

```
> Equations := Equations_xy union Equations_tx mod p:

  Nombre_Equations := nops(Equations);
```

$$Nombre_Equations := 35378 \quad (2)$$

```
> Variables :=     {seq(seq(seq(cR[m-0-j,j,dR-q-r,q,r], q=0..dR-
  r), r=0..dR), j=0..m-0)}
           union {seq(seq(seq(cS[m-3-j,j,dS-q-r,q,r], q=0..dS-
  r), r=0..dS), j=0..m-3)}
           union {seq(seq(seq(cU[m-6-j,j,dU-q-r,q,r], q=0..dU-
  r), r=0..dU), j=0..m-6)}
           union {seq(seq(seq(cV[m-9-j,j,dV-q-r,q,r], q=0..dV-
  r), r=0..dV), j=0..m-9)}
           union {seq(seq(seq(cW[m-12-j,j,dW-q-r,q,r], q=0..dW-
  r), r=0..dW), j=0..m-12)}:

  Nombre_Variables := nops(Variables);
```

$$Nombre_Variables := 12550 \quad (3)$$

```
> Solutions := msolve(Equations,p):
```

```
> Verification := seq(op(2,op(i,Solutions)),i=1..nops(Solutions));
  indets({Verification}):
  nops(indets({Verification})):
```

$$Verification := 0, 0, 0,

0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,

0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,

0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,

0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,

0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,

0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,

0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,

0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,

0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,

0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,

0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0

```
> assign(Solutions):
```

```
> for j from 0 to m-0 do
  Voir_R[m-0-j,j] := add(add(cR[m-0-j,j,dR-q-r,q,r]*T^(dR-q-r)*X^q*
  Y^r, q=0..dR-r), r=0..dR):
  od;

  for j from 0 to m-3 do
  Voir_S[m-3-j,j] := add(add(cS[m-3-j,j,dS-q-r,q,r]*T^(dS-q-r)*X^q*
  Y^r, q=0..dS-r), r=0..dS):
  od;

  for j from 0 to m-6 do
  Voir_U[m-6-j,j] := add(add(cU[m-6-j,j,dU-q-r,q,r]*T^(dU-q-r)*X^q*
  Y^r, q=0..dU-r), r=0..dU):
  od;

  for j from 0 to m-9 do
  Voir_V[m-9-j,j] := add(add(cV[m-9-j,j,dV-q-r,q,r]*T^(dV-q-r)*X^q*
  Y^r, q=0..dV-r), r=0..dV):
  od;

  for j from 0 to m-12 do
  Voir_W[m-12-j,j] := add(add(cW[m-12-j,j,dW-q-r,q,r]*T^(dW-q-r)*
  X^q*Y^r, q=0..dW-r), r=0..dW):
  od;
```

$$Voir_R_{13,\,0} := 0$$
$$Voir_R_{12,\,1} := 0$$
$$Voir_R_{11,\,2} := 0$$
$$Voir_R_{10,\,3} := 0$$
$$Voir_R_{9,\,4} := 0$$
$$Voir_R_{8,\,5} := 0$$
$$Voir_R_{7,\,6} := 0$$
$$Voir_R_{6,\,7} := 0$$

$$Voir_R_{5,\,8} := 0$$
$$Voir_R_{4,\,9} := 0$$
$$Voir_R_{3,\,10} := 0$$
$$Voir_R_{2,\,11} := 0$$
$$Voir_R_{1,\,12} := 0$$
$$Voir_R_{0,\,13} := 0$$
$$Voir_S_{10,\,0} := 0$$
$$Voir_S_{9,\,1} := 0$$
$$Voir_S_{8,\,2} := 0$$
$$Voir_S_{7,\,3} := 0$$
$$Voir_S_{6,\,4} := 0$$
$$Voir_S_{5,\,5} := 0$$
$$Voir_S_{4,\,6} := 0$$
$$Voir_S_{3,\,7} := 0$$
$$Voir_S_{2,\,8} := 0$$
$$Voir_S_{1,\,9} := 0$$
$$Voir_S_{0,\,10} := 0$$
$$Voir_U_{7,\,0} := 0$$
$$Voir_U_{6,\,1} := 0$$
$$Voir_U_{5,\,2} := 0$$
$$Voir_U_{4,\,3} := 0$$
$$Voir_U_{3,\,4} := 0$$
$$Voir_U_{2,\,5} := 0$$
$$Voir_U_{1,\,6} := 0$$
$$Voir_U_{0,\,7} := 0$$
$$Voir_V_{4,\,0} := 0$$
$$Voir_V_{3,\,1} := 0$$
$$Voir_V_{2,\,2} := 0$$
$$Voir_V_{1,\,3} := 0$$
$$Voir_V_{0,\,4} := 0$$
$$Voir_W_{1,\,0} := 0$$
$$Voir_W_{0,\,1} := 0 \qquad \textbf{(5)}$$